P2 Chapter 11: Integration

Using Reverse Chain Rule

SKILL #4: Reverse Chain Rule

✏ **Integration by Inspection/Reverse Chain Rule**: Use common sense to **consider some expression** that would differentiate to the expression given. Then **scale** appropriately.
Common patterns:

$$\int k\frac{f'(x)}{f(x)}dx \rightarrow \quad Try\ \ln|f(x)|$$

In words: "If the bottom of a fraction differentiates to give the top (forgetting scaling), try ln of the bottom".

$$\int kf'(x)[f(x)]^n \rightarrow \quad Try\ [f(x)]^{n+1}$$

$$\int \frac{x^2}{x^3+1}\,dx$$

Consider $y = \ln|x^3+1|$

$$\frac{dy}{dx} = \frac{3x^2}{x^3+1}$$

Therefore:

$$\int \frac{x^2}{x^3+1}\,dx$$

$$= \frac{1}{3}ln|x^3+1| + C$$

$$\int x\,e^{x^2+1}\,dx$$

Consider $y = e^{x^2+1}$

$$\frac{dy}{dx} = 2x\,e^{x^2+1}$$

Therefore:

$$\int x\,e^{x^2+1}dx = \frac{1}{2}e^{x^2+1} + C$$

Quickfire

In your head !

$$\int \frac{4x^3}{x^4 - 1}\,dx = \boxed{?}$$

$$\int \frac{\cos x}{\sin x + 2}\,dx = \boxed{?}$$

$$\int \cos x \; e^{\sin x}\,dx = \boxed{?}$$

$$\int \cos x\,(\sin x - 5)^7\,dx = \boxed{?}$$

$$\int x^2(x^3 + 5)^7\,dx = \boxed{?}$$

Not in your head !?

$$\int \frac{x}{(x^2 + 5)^3}\,dx = \boxed{?}$$

Quickfire

In your head !

$$\int \frac{4x^3}{x^4-1}\,dx = \mathbf{ln}\left|\boldsymbol{x^4}-\mathbf{1}\right| + \boldsymbol{C}$$

$$\int \frac{\cos x}{\sin x + 2}\,dx = \boldsymbol{ln}|\mathbf{sin}\,\boldsymbol{x}+\mathbf{2}| + \boldsymbol{C}$$

$$\int \cos x\ \ e^{\sin x}\,dx = \boldsymbol{e}^{\mathbf{sin}\,\boldsymbol{x}} + \boldsymbol{C}$$

$$\int \cos x\,(\sin x - 5)^7\,dx = \frac{\mathbf{1}}{\mathbf{8}}(\mathbf{sin}\,\boldsymbol{x}-\mathbf{5})^{\mathbf{8}} + \boldsymbol{C}$$

$$\int x^2(x^3+5)^7\,dx = \frac{\mathbf{1}}{\mathbf{24}}\left(\boldsymbol{x^3}+\mathbf{5}\right)^{\mathbf{8}} + \boldsymbol{C}$$

Not in your head !?

$$\int \frac{x}{(x^2+5)^3}\,dx = \int \boldsymbol{x}\left(\boldsymbol{x^2}+\mathbf{5}\right)^{-\mathbf{3}}\boldsymbol{dx}$$

$$-\frac{\mathbf{1}}{\mathbf{4}}\left(\boldsymbol{x^2}+\mathbf{5}\right)^{-\mathbf{2}} + \boldsymbol{C}$$

$\sin^n x \cos x$ vs $\sec^n x \tan x$

Notice when we differentiate $\sin^5 x$, then power decreases:

$$\frac{d}{dx}\left(\sin^5 x\right) = \frac{d}{dx}\left((\sin x)^5\right) = 5\sin^4 x \cos x$$

However, when we differentiate $\sec^5 x$:

$$\frac{d}{dx}\left((\sec x)^5\right) = 5\sec^4 x \times \sec x \tan x = 5\sec^5 x \tan x$$

Notice that the power of *sec* didn't go down. Keep this in mind when integrating.

$$\int \sec^4 x \tan x \ dx$$

Consider $y = \sec^4 x$

$$\frac{dy}{dx} = 4\sec^3 x \times \sec x \tan x = 4\sec^4 x \tan x$$

$$\therefore \int sec^4 x \, tan \, x \ dx = \frac{1}{4}\sec^4 x + C$$

Test Your Understanding

$$\int \sin x\,(\cos x + 1)^5\,dx$$

?

$$\int \frac{cosec^2 x}{(2 + \cot x)^3}\,dx$$

?

$$\int \frac{\sec^2 2x}{\tan 2x + 1}\,dx$$

?

$$\int x(x^2 + 2)^3\,dx$$

?

$$\int 5\tan x \sec^2 x\;dx$$

?

Test Your Understanding

$$\int \sin x\,(\cos x + 1)^5\,dx$$

Consider $y = (\cos x + 1)^6$

$$\frac{dy}{dx} = -6\sin x\,(\cos x + 1)^5$$

Therefore:

$$\int \sin x\,(\cos x + 1)^5\,dx$$

$$= -\frac{\mathbf{1}}{\mathbf{6}}(\mathbf{\cos x + 1})^{\mathbf{6}} + \boldsymbol{C}$$

$$\int \frac{cosec^2 x}{(2 + \cot x)^3}\,dx$$

Consider $y = (2 + \cot x)^{-2}$

$$\frac{dy}{dx} = -2(2 + \cot x)^{-3} \times -cosec^2 x$$

$$= 2cosec^2 x(2 + \cot x)^{-3}$$

Thus $\int \frac{cosec^2 x}{(2+\cot x)^3}\,dx = \frac{\mathbf{1}}{\mathbf{2}}(\mathbf{2} + \mathbf{\cot x})^{-\mathbf{2}} + \boldsymbol{C}$

$$\int \frac{\sec^2 2x}{\tan 2x + 1}\,dx$$

$$= \frac{1}{2}\ln|\tan 2x + 1| + C$$

$$\int x(x^2 + 2)^3\,dx$$

$$= \frac{1}{8}(x^2 + 2)^4 + C$$

$$\int 5\tan x \sec^2 x\;dx$$

Try $y = \sec^2 x$

$$\frac{dy}{dx} = 2\sec x \times \sec x \tan x$$

$$= 2\sec^2 x \tan x$$

$$\therefore \int 5\tan x \sec^2 x\;dx = \frac{5}{2}\sec^2 x + C$$

Exercise 11.4

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Extension:

[STEP I 2013 Q4]
(i) Show that, for $n > 0$,
$\int_0^{\frac{1}{4}\pi} \tan^n x \sec^2 x \ dx = \frac{1}{n+1}$ and $\int_0^{\frac{1}{4}\pi} \sec^n x \tan x \ dx = \frac{\left(\sqrt{2}\right)^n - 1}{n}$
(ii) Evaluate the following integrals:
$\int_0^{\frac{1}{4}\pi} x \sec^4 x \tan x \ dx$ and $\int_0^{\frac{1}{4}\pi} x^2 \sec^2 x \tan x \ dx$

Solutions: (ii) ?

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[STEP I 2013 Q4]
(i) Show that, for $n > 0$,
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(ii) Evaluate the following integrals:
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Solutions: (ii) $\boldsymbol{\frac{\pi}{4} - \frac{1}{3}}$ **and** $\boldsymbol{\frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2}$

Homework Exercise

Hint Decide carefully whether each expression is in the form $k\frac{f'(x)}{f(x)}$ or $kf'(x)(f(x))^n$.

1 Integrate the following functions.

a $\frac{x}{x^2+4}$ **b** $\frac{e^{2x}}{e^{2x}+1}$ **c** $\frac{x}{(x^2+4)^3}$

d $\frac{e^{2x}}{(e^{2x}+1)^3}$ **e** $\frac{\cos 2x}{3+\sin 2x}$ **f** $\frac{\sin 2x}{(3+\cos 2x)^3}$

g xe^{x^2} **h** $\cos 2x(1+\sin 2x)^4$ **i** $\sec^2 x \tan^2 x$ **j** $\sec^2 x(1+\tan^2 x)$

2 Find the following integrals.

a $\int (x+1)(x^2+2x+3)^4\,dx$ **b** $\int \operatorname{cosec}^2 2x \cot 2x\,dx$

c $\int \sin^5 3x \cos 3x\,dx$ **d** $\int \cos x\, e^{\sin x}\,dx$

e $\int \frac{e^{2x}}{e^{2x}+3}\,dx$ **f** $\int x(x^2+1)^{\frac{3}{2}}\,dx$

g $\int (2x+1)\sqrt{x^2+x+5}\,dx$ **h** $\int \frac{2x+1}{\sqrt{x^2+x+5}}\,dx$

i $\int \frac{\sin x \cos x}{\sqrt{\cos 2x+3}}\,dx$ **j** $\int \frac{\sin x \cos x}{\cos 2x+3}\,dx$

3 Find the exact value of each of the following:

a $\int_0^3 (3x^2+10x)\sqrt{x^3+5x^2+9}\,dx$ **b** $\int_{\frac{\pi}{9}}^{\frac{2\pi}{9}} \frac{6\sin 3x}{1-\cos 3x}\,dx$

c $\int_4^7 \frac{x}{x^2-1}\,dx$ **d** $\int_0^{\frac{\pi}{4}} \sec^2 x\, e^{4\tan x}\,dx$

4 Given that $\int_0^k kx^2e^{x^3}dx = \frac{2}{3}(e^8-1)$, find the value of k. **(3 marks)**

5 Given that $\int_0^\theta 4\sin 2x \cos^4 2x\,dx = \frac{4}{5}$ where $0 < \theta < \pi$, find the exact value of θ.

6 **a** By writing $\cot x = \frac{\cos x}{\sin x}$, find $\int \cot x\,dx$. **(2 marks)**

b Show that $\int \tan x\,dx \equiv \ln|\sec x| + c$. **(3 marks)**

Homework Answers

1 **a** $\frac{1}{2}\ln|x^2+4|+c$ **b** $\frac{1}{2}\ln|e^{2x}+1|+c$

c $-\frac{1}{4}(x^2+4)^{-2}+c$ **d** $-\frac{1}{4}(e^{2x}+1)^{-2}+c$

e $\frac{1}{2}\ln|3+\sin 2x|+c$ **f** $\frac{1}{4}(3+\cos 2x)^{-2}+c$

g $\frac{1}{2}e^{x^2}+c$ **h** $\frac{1}{10}(1+\sin 2x)^5+c$

i $\frac{1}{3}\tan^3 x+c$ **j** $\tan x+\frac{1}{3}\tan^3 x+c$

2 **a** $\frac{1}{10}(x^2+2x+3)^5+c$ **b** $-\frac{1}{4}\cot^2 2x+c$

c $\frac{1}{18}\sin^6 3x+c$ **d** $e^{\sin x}+c$

e $\frac{1}{2}\ln|e^{2x}+3|+c$ **f** $\frac{1}{5}(x^2+1)^{\frac{5}{2}}+c$

g $\frac{2}{3}(x^2+x+5)^{\frac{3}{2}}+c$ **h** $2(x^2+x+5)^{\frac{1}{2}}+c$

i $-\frac{1}{2}(\cos 2x+3)^{\frac{1}{2}}+c$ **j** $-\frac{1}{4}\ln|\cos 2x+3|+c$

3 **a** 468 **b** $2\ln 3$ **c** $\frac{1}{2}\ln\left(\frac{16}{5}\right)$ **d** $\frac{1}{4}(e^4-1)$

4 $k=2$ **5** $\theta=\frac{\pi}{2}$

6 **a** $\ln|\sin x|+c$

b $\int \tan x\,dx = -\ln|\cos x|+c = \ln\left|\frac{1}{\cos x}\right|+c$

$= \ln|\sec x|+c$